SignLab Write-Up

This app is a small project, but I think it shows a really big idea. SignLab lets a person practice the ASL alphabet with a webcam. The camera looks at a hand, the app finds hand points, and then it makes a guess about which letter the person is trying to sign. What I like about this project is that it feels interactive. Instead of just looking at a poster of handshapes, the user gets to practice, save examples, and see a live guess on the screen.

I improved the app in a few ways that make it feel more complete and easier to use. First, I cleaned up the design so it looks nicer and is easier to understand. Second, I changed the single big alphabet chart into separate images for each letter, which makes the reference section much more useful. Third, I added all 26 letters, a way to reset just one letter at a time, and localStorage so the user's progress stays saved even after refreshing the page. I also added a word generator so people can practice a full word instead of only one letter at a time.

Machine vision can be really helpful for accessibility and communication. A tool like this could help beginners practice fingerspelling on their own and build confidence. More broadly, machine vision can support people by reading text aloud, helping identify objects, or making technology easier to use without a keyboard or mouse. For communication, it could be part of tools that help people learn sign language or interact in more inclusive ways. It is important to remember, though, that this app is only a simple learning tool. Real sign language is much richer than a single handshape. It includes movement, facial expression, and grammar.

At the same time, machine vision can also be used in harmful ways. Cameras can be used for constant surveillance, which can make people feel watched and take away privacy. Data from cameras can also be stored, shared, or used without a person's clear permission. Another problem is bias: if a system is trained poorly, it may work better for some people than others. Machine vision can also be used in military systems, where mistakes could have very serious consequences. Because of that, I think tools like this should be built carefully, with honesty about their limits and respect for the people being affected.

Overall, SignLab is a simple example of how machine vision can be used for something positive. It helps with practice, learning, and accessibility in a way that feels friendly and approachable. At the same time, it also reminds us that technology is not automatically good just because it is impressive. We still have to think about privacy, fairness, and how these systems affect real people.